

# CS5691: Pattern Recognition and Machine Learning

## Programming Assignment 3

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# 1 Task 1: Function Approximation for Dataset 1 using MLFFNN with One Hidden Layer

## 1.1 Introduction

In this task, we implement a Multilayer Feedforward Neural Network (MLFFNN) with one hidden layer containing 8 nodes. The objective is to perform function approximation on Dataset 1.

## 1.2 Training and Results

The training process involves running the MLFFNN for a sufficient number of epochs until convergence. The following figures represent the results:

### 1.2.1 Training Error vs Epoch Plot

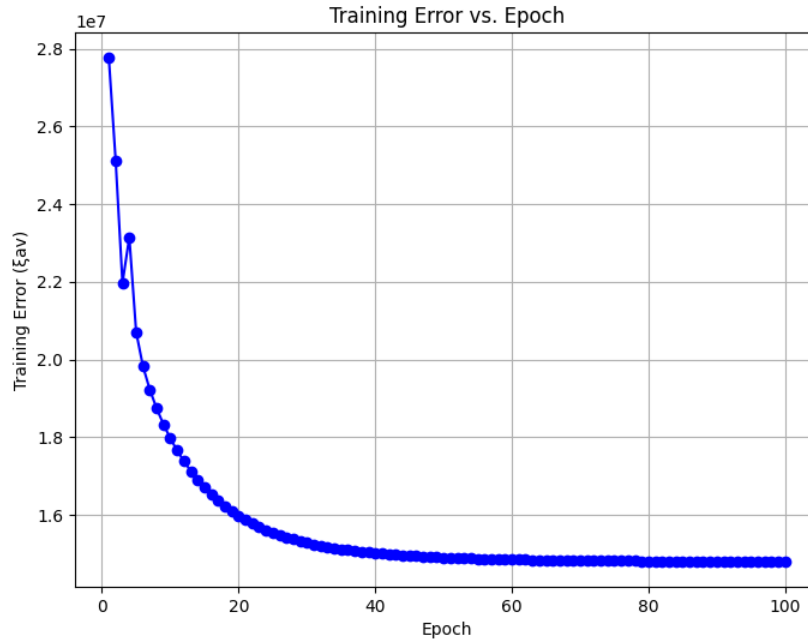


Figure 1: Training Error ( $\xi_{av}$ ) vs Epoch for Dataset 1

### 1.2.2 Scatter Plots

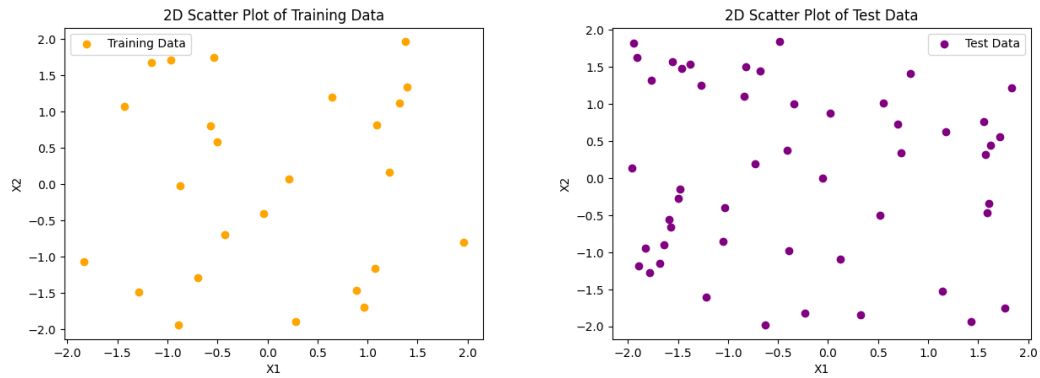


Figure 2: Scatter Plots: Left - Training Data, Right - Test Data

### 1.2.3 Surface Plots

We visualize the output of two nodes from the hidden layer and one node from the output layer at different epochs (1, 10, 50, and at convergence):

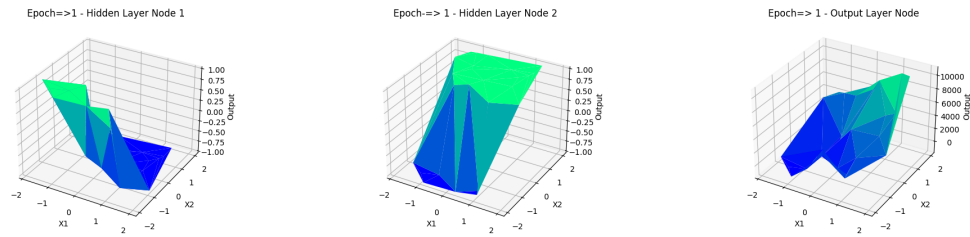


Figure 3: Surface Plots at Epoch 1

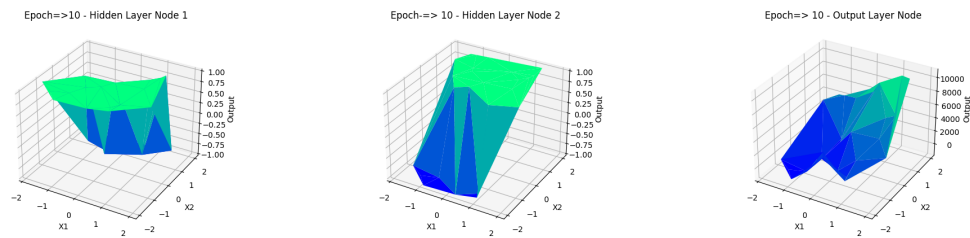


Figure 4: Surface Plots at Epoch 10

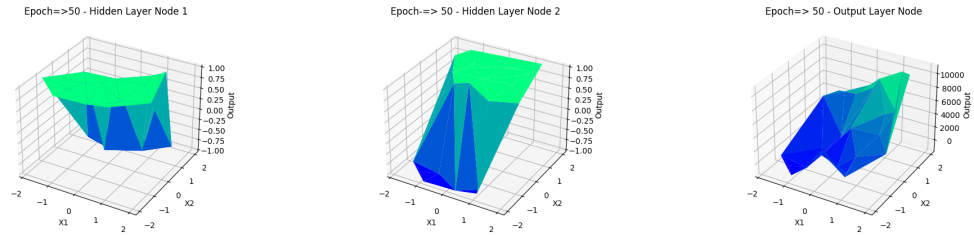


Figure 5: Surface Plots at Epoch 50

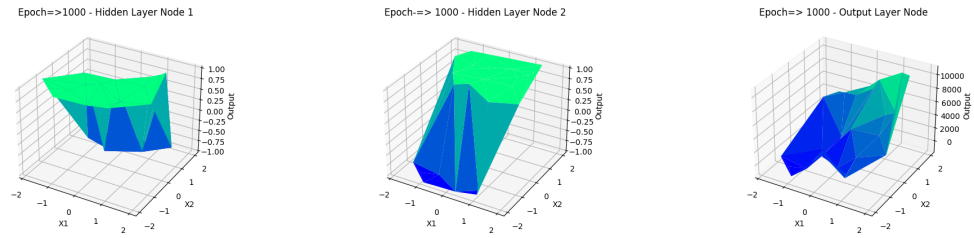


Figure 6: Surface Plots at Convergence

### 1.3 Summary

In this task, we successfully implemented an MLFFNN with one hidden layer for function approximation on Dataset 1. The model demonstrated effective learning, as shown by the training error reduction and the surface plots, which exhibit changes in the hidden and output nodes over time. The scatter plots of the training and test data show how well the model approximated the function for both datasets.

## 2 Task 2: Function Approximation for Dataset 2 using MLFFNN with Two Hidden Layers

### 2.1 Introduction

In this task, we implement a Multilayer Feedforward Neural Network (MLFFNN) with two hidden layers. The first hidden layer contains 15 nodes, and the second hidden layer contains 10 nodes. The objective is to perform function approximation on Dataset 2.

### 2.2 Training and Results

The training process involves running the MLFFNN for a sufficient number of epochs until convergence. The following figures represent the results:

#### 2.2.1 Training Error vs Epoch Plot

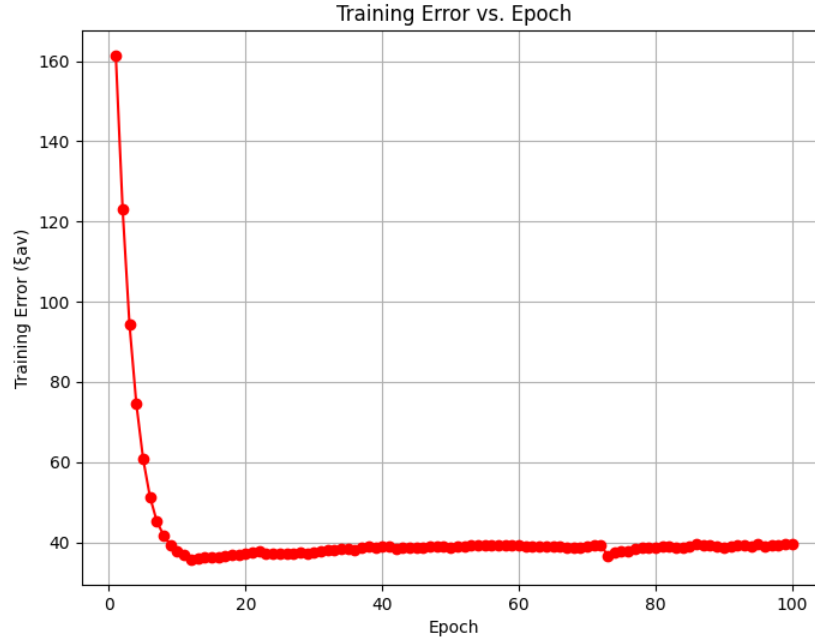


Figure 7: Training Error ( $\xi_{av}$ ) vs Epoch for Dataset 2

### 2.2.2 Scatter Plots

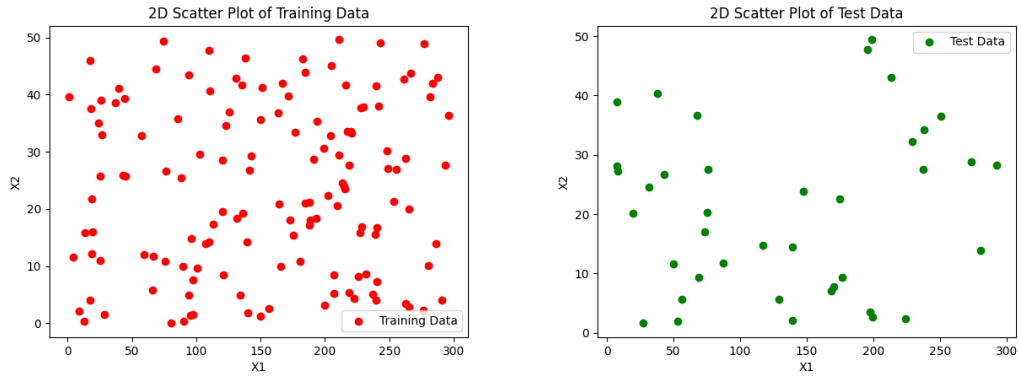


Figure 8: Scatter Plots: Left - Training Data, Right - Test Data

## 2.3 Summary

In this task, we successfully implemented an MLFFNN with two hidden layers for function approximation on Dataset 2. The model demonstrated effective learning, as shown by the training error reduction and the surface plots, which exhibit changes in the hidden and output nodes over time. The scatter plots of the training and test data show how well the model approximated the function for both datasets.

### 3 Task 3: Classifiers for Dataset 3

#### 3.1 Introduction

In this task, we explore two classification models for Dataset 3: (a) a GMM-based classifier with 8 Gaussians per class, and (b) an MLFFNN-based classifier with two hidden layers (12 nodes in the first hidden layer and 8 nodes in the second hidden layer). The goal is to assess classification performance through confusion matrices, classification accuracy, decision region plots, and surface plots for the MLFFNN.

#### 3.2 GMM-Based Classifier

The GMM-based classifier is evaluated with both (i) full covariance matrices and (ii) diagonal covariance matrices. The GMM classifies the data based on the distributional characteristics of each class using Gaussian mixtures.

##### 3.2.1 Confusion Matrices and Classification Accuracies

We present the confusion matrices and classification accuracies for both the training and test data for the two configurations of the GMM-based classifier:

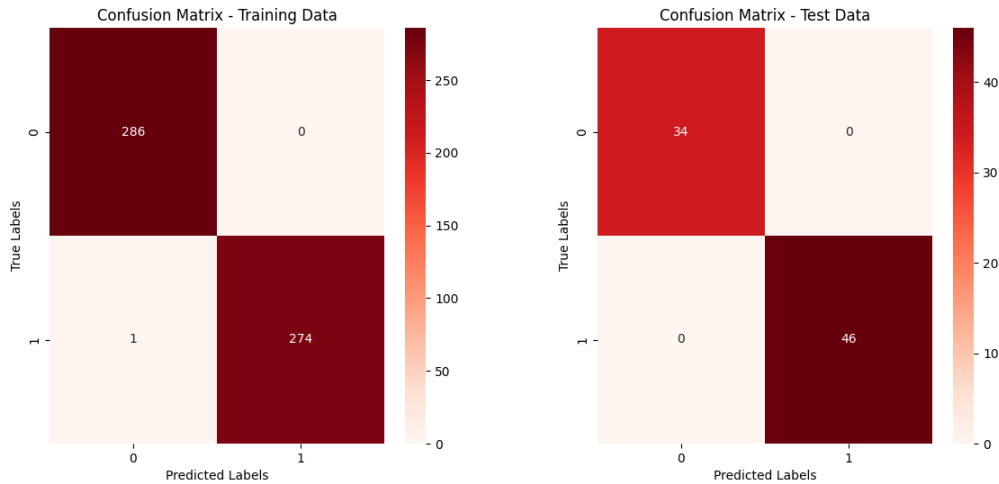


Figure 9: Confusion Matrix: Left - Full Covariance (Train), Right - Full Covariance (Test)

- Training Accuracy: 99.82%
- Test Accuracy: 100%

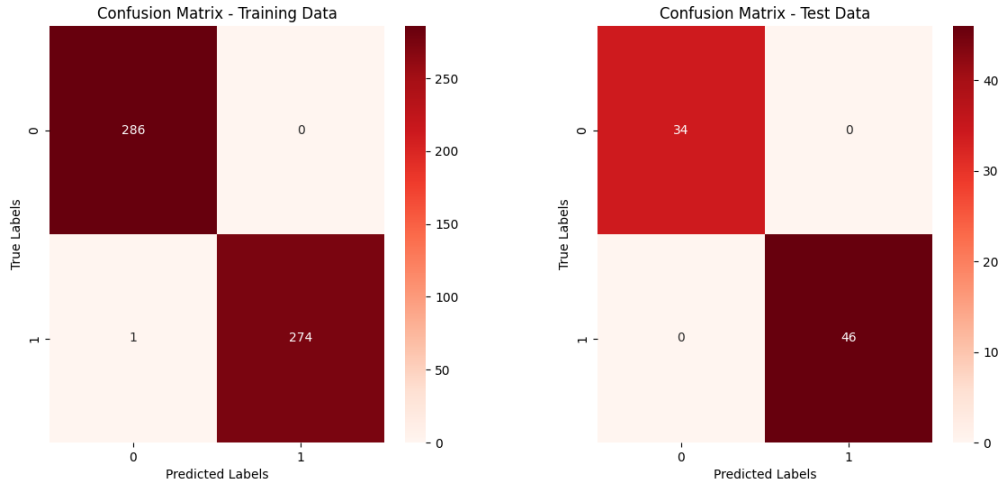


Figure 10: Confusion Matrix: Left - Diagonal Covariance (Train), Right - Diagonal Covariance (Test)

- Training Accuracy: 99.91%
- Test Accuracy: 100%

### 3.2.2 Decision Region Plots

The following plots illustrate the decision regions generated by the GMM-based classifier for both the full and diagonal covariance configurations, superposed with the training data and the level curves of the Gaussians.

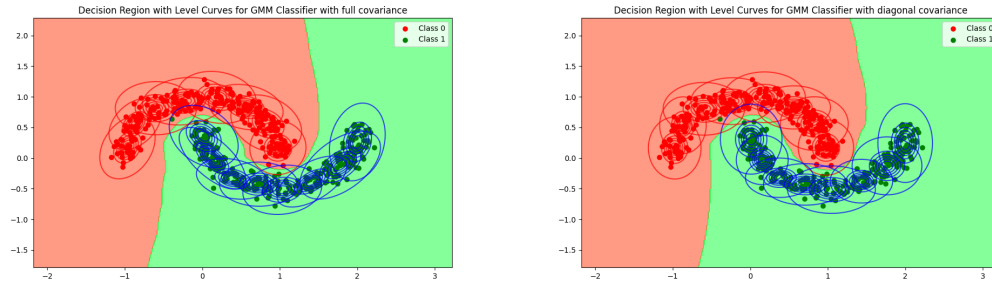


Figure 11: Decision Region Plots: Left - Full Covariance, Right - Diagonal Covariance

## 3.3 MLFFNN-Based Classifier

The MLFFNN classifier uses two hidden layers with 12 nodes in the first hidden layer and 8 nodes in the second hidden layer. The activation function in the hidden layers is the Tanh function, and the output layer uses the softmax function for classification tasks. Cross-entropy is used as the error function.

### 3.3.1 Confusion Matrices and Classification Accuracies

The confusion matrices and classification accuracies for the MLFFNN classifier on both the training and test datasets are presented below:



Figure 12: Confusion Matrix: Left - Train, Right - Test (MLFFNN)

- Training Accuracy: 99.82%
- Test Accuracy: 98.75%

### 3.3.2 Decision Region Plot

The decision region plot for the MLFFNN classifier is shown below, superposed with the training data:

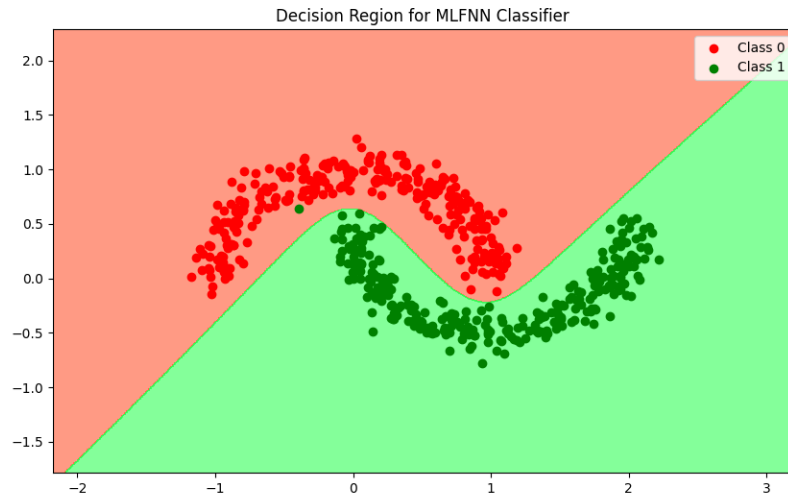


Figure 13: Decision Region Plot for MLFFNN Classifier (Training Data Superposed)

### 3.3.3 Surface Plots

We visualize the outputs of two nodes from the first hidden layer, two nodes from the second hidden layer, and the output nodes, at different epochs (1, 10, 50, and at convergence). Each epoch consists of

6 surface plots—two for each layer.  
**Epoch 1:**

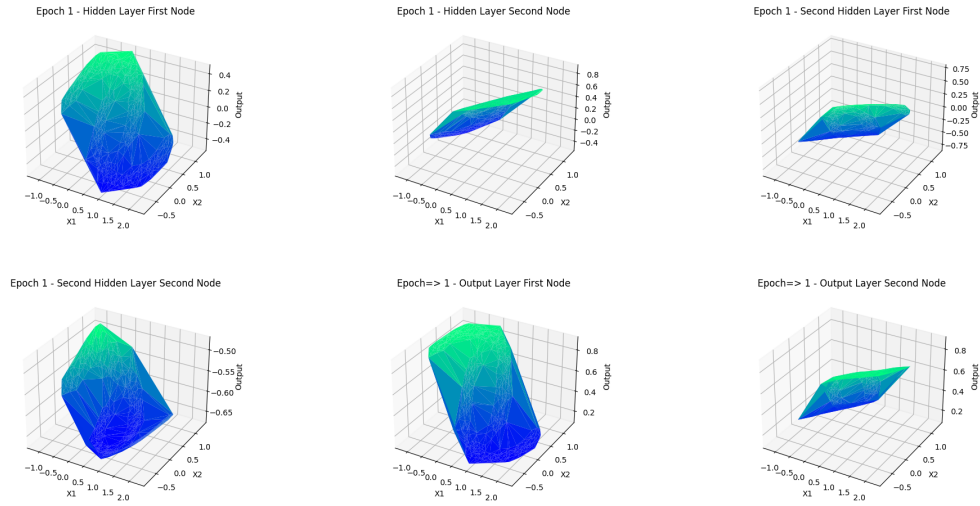


Figure 14: Surface Plots at Epoch 1

**Epoch 10:**

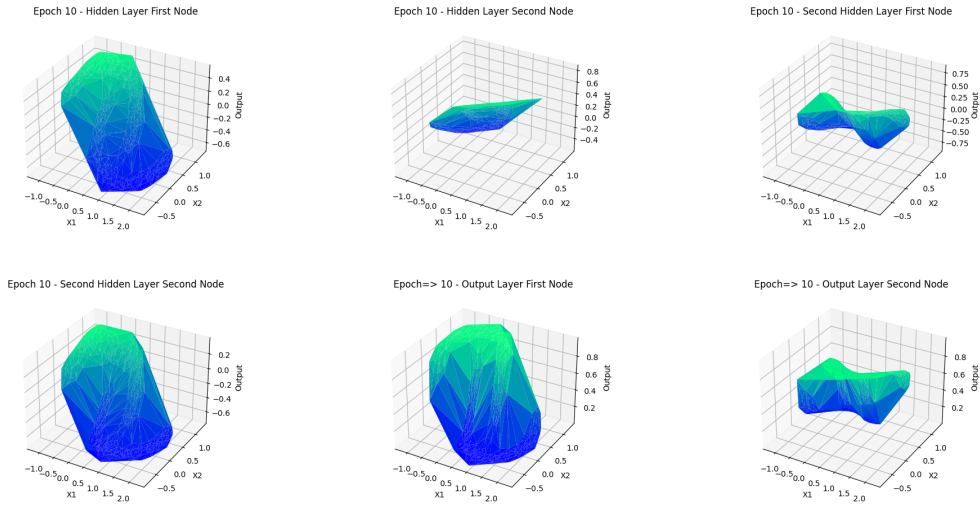


Figure 15: Surface Plots at Epoch 10

**Epoch 50:**

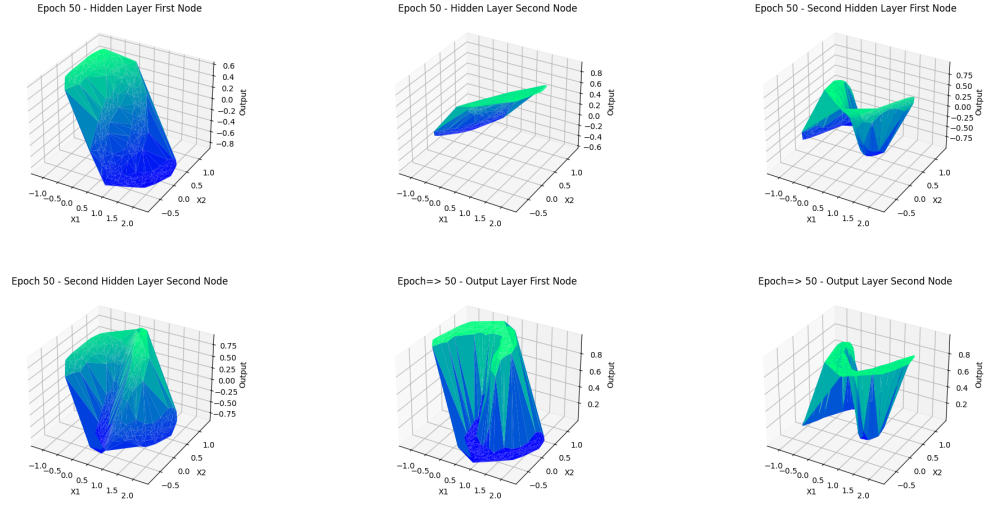


Figure 16: Surface Plots at Epoch 50

**At Convergence:**

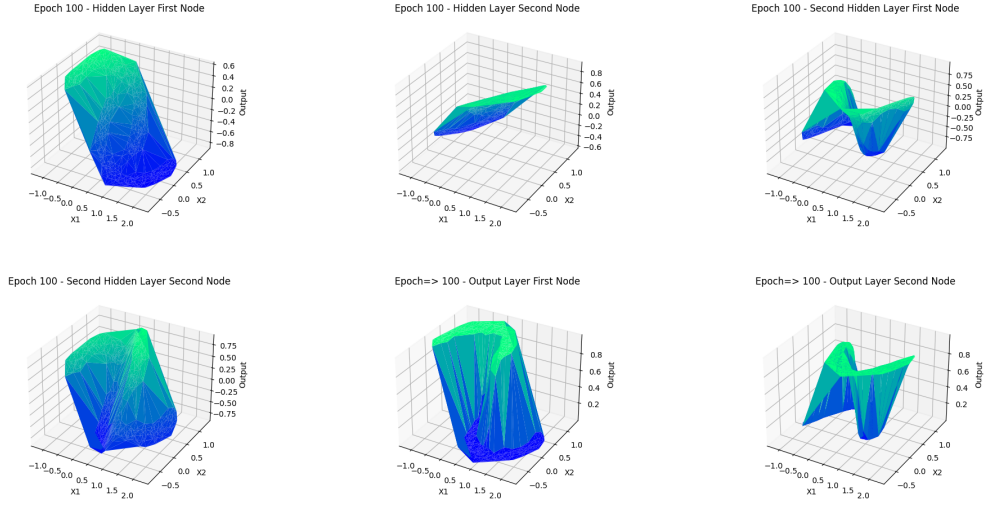


Figure 17: Surface Plots at Convergence

### 3.3.4 Training Error vs Epoch Plot

Finally, we present the plot of the average training error  $\xi_{av}$  vs epoch for the MLFFNN classifier:

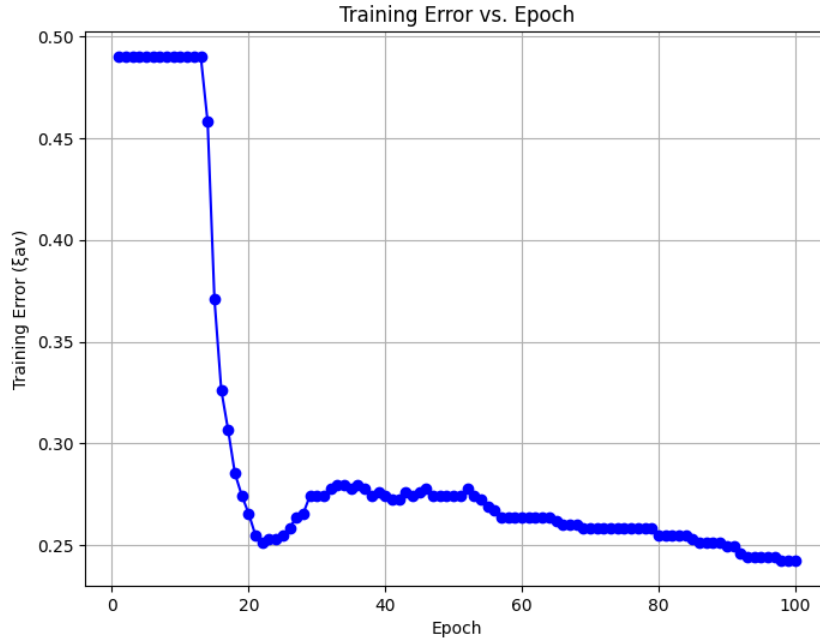


Figure 18: Training Error ( $\xi_{av}$ ) vs Epoch for Dataset 3 (MLFFNN)

### 3.4 Summary

In this task, we successfully implemented and evaluated two classifiers for Dataset 3: a GMM-based classifier and an MLFFNN-based classifier. The confusion matrices, decision region plots, surface plots, and training error plots provide a comprehensive analysis of the performance and learning dynamics of the models.

## 4 Task 4: Classifiers for Dataset 4

### 4.1 Introduction

In this task, we evaluate two classification models for Dataset 4: (a) a GMM-based classifier with 3 Gaussians per class, and (b) an MLFFNN-based classifier with two hidden layers (25 nodes in the first hidden layer and 15 nodes in the second hidden layer). The objective is to assess classification performance through confusion matrices, classification accuracy, and training error plots.

### 4.2 GMM-Based Classifier

The GMM-based classifier is evaluated with both (i) full covariance matrices and (ii) diagonal covariance matrices. The GMM classifies the data by modeling the distribution of each class with a mixture of Gaussians.

#### 4.2.1 Confusion Matrices and Classification Accuracies

The confusion matrices and classification accuracies for the GMM-based classifier on both the training and test datasets for full and diagonal covariance matrices are presented below:

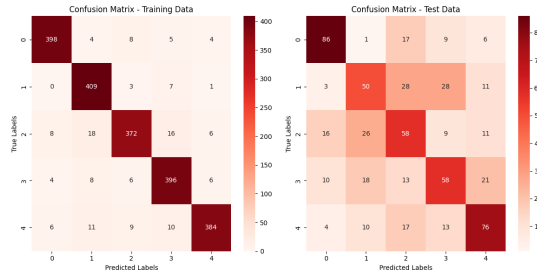


Figure 19: Confusion Matrix: Left - Full Covariance (Train), Right - Full Covariance (Test)

- Training Accuracy: 93.33%
- Test Accuracy: 54.76%

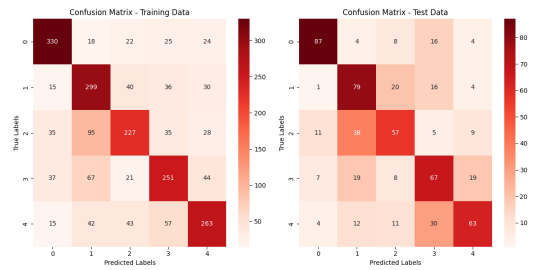


Figure 20: Confusion Matrix: Left - Diagonal Covariance (Train), Right - Diagonal Covariance (Test)

- Training Accuracy: 65.27%
- Test Accuracy: 58.93%

### 4.3 MLFFNN-Based Classifier

The MLFFNN classifier uses two hidden layers with 25 nodes in the first hidden layer and 15 nodes in the second hidden layer. The activation function for the hidden layers is the Tanh function, and the output layer uses the softmax function for classification. The cross-entropy function is used as the error function.

#### 4.3.1 Confusion Matrices and Classification Accuracies

The confusion matrices and classification accuracies for the MLFFNN classifier on both the training and test datasets are shown below:

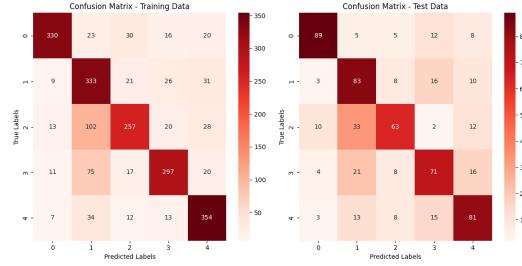


Figure 21: Confusion Matrix: Left - Train, Right - Test (MLFFNN)

- Training Accuracy: 74.85%
- Test Accuracy: 64.61%

#### 4.3.2 Training Error vs Epoch Plot

The plot of the average training error  $\xi_{av}$  vs epoch for the MLFFNN classifier is presented below:

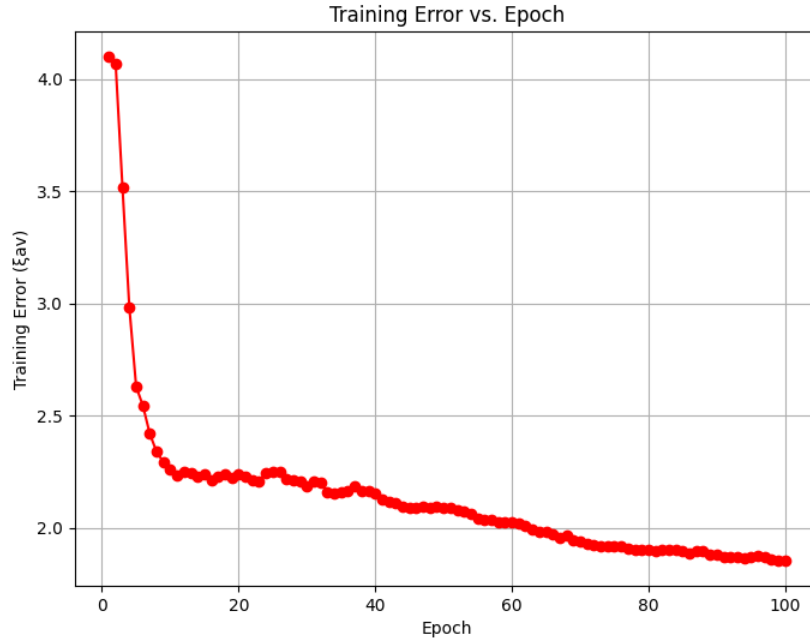


Figure 22: Training Error ( $\xi_{av}$ ) vs Epoch for Dataset 4 (MLFFNN)

#### 4.4 Summary

In Task 4, we successfully implemented and evaluated two classifiers for Dataset 4: a GMM-based classifier and an MLFFNN-based classifier. The confusion matrices and training error plots provided insights into the performance of each classifier. Both models showed different behavior in terms of classification accuracy and error convergence, with the MLFFNN classifier demonstrating effective learning through its training error reduction.